

Take Control of Censoring, Generate Real-World Like Synthetic Data

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Abstract

Benchmarking survival models under varying censoring regimes requires data where censoring is both realistic and controllable. We propose a semi-synthetic framework that, starting from a real survival dataset, generates derived datasets with an *exact* user-specified censoring rate among initially uncensored subjects, through covariate-informed selection of subjects to censor and subject-specific censoring-time sampling. We evaluate realism with Wasserstein distance to a reproducible reference distribution and assess preservation of event-censoring dependence using copula-based Kendall's τ on synthetic settings with observable (T, C) . Across six public datasets and censoring rates from 10% to 90%, our method better matches the reference distribution and preserves dependence compared to parametric and random baselines. Code is available at <https://anonymous.4open.science/r/Semi-Synthetic-Censoring-Simulation-FDB7/README.md>.

CCS Concepts

• **Mathematics of computing** → **Survival analysis**.

Keywords

survival analysis, censored data, synthetic data generation, machine learning, structured data, statistical dependence

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1 Introduction

Survival analysis deals with a wide range of empirical comparisons in data mining, where methods are routinely benchmarked on time-to-event prediction tasks such as disease onset [19], customer churn

[15], mechanical system failure [19], and other endpoints of interest [25]. A central obstacle to reliable benchmarking is right censoring: the event time is only partially observed when it does not occur during the study period or becomes unobservable [2, 21]. Crucially, censoring acts as a domain shift that changes the observed data distribution and can vary dramatically across application areas, from typical 20%–40% rates in medical studies to more than 90% in aerospace settings. Such variability breaks comparability across datasets and studies, and it complicates controlled experiments (e.g., ablations, stress-tests, and robustness curves). Both the amount and the mechanism of missingness directly alter which information is available to each method and to the evaluator. As a result, apparent improvements on one benchmark may reflect differences in censoring rather than genuine modeling progress, and model rankings can become fragile when censoring is heavy or informative. This motivates the need for benchmarks where censoring can be systematically controlled while keeping the underlying event process fixed. Moreover, common evaluation criteria are affected unevenly. For instance, pointwise error metrics such as the mean absolute error (MAE) require individual event times and therefore cannot be directly computed for observations whose true event time is unobserved due to censoring [20].

Real datasets rarely offer the censoring rate or mechanism required for a targeted study, making controlled censoring generation key to reliable evaluation. Such controlled generation would enable benchmarks where only censoring varies, support fair comparisons and ranking stability analyses, and extend experiments to extreme-censoring regimes often missing from public datasets. This need is well recognized in the literature: [27] generate data with predefined censoring rates (under proportional hazards) to study treatment effect estimation; [12] use Monte Carlo simulations on real datasets to quantify how censoring biases comparative effectiveness studies; [22] introduce tools to generate realistic censored survival-time distributions for clinical trial design; and [26, 28] rely on simulation to study sensitivity analyses and decision-making under censoring.

Traditional simulation pipelines typically generate covariates, event times T , and censoring times C from predefined statistical models [26]. Event times are often drawn from simple distributions (e.g., exponential, Weibull) or from parametric regression models [23]; censoring times are then sampled from another distribution and combined through the observed time $Y = \min(T, C)$. A practical advantage of these parametric constructions is that the

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global censoring rate can be matched by tuning distribution parameters via integration or root-finding [16]. However, this control relies on strong assumptions: real censoring times do not necessarily follow convenient families, and rate control is most straightforward when censoring is generated independently of covariates. Bootstrap resampling can also adjust censoring proportions [3], but it introduces duplicates that may distort the original data structure and reduce variability.

To better reflect realistic dependence patterns, especially under dependent censoring, several lines of work aim to jointly model event and censoring times. Copula-based approaches explicitly model (T, C) and allow control over the strength and structure of dependence [8], enabling more realistic synthetic datasets [9, 29]. More recently, generative approaches such as SurvivalGAN [17] and augmentation strategies like [30] learn complex joint distributions of covariates, events, and censoring for data generation or enrichment. Despite their ability to capture complex dependencies, these approaches rarely offer precise, predefined control of the *global* censoring rate; moreover, copula methods hinge on parametric choices that may be misspecified. Generative models typically target a different objective—high-fidelity synthetic data generation/enrichment (often for privacy-sensitive settings)—and therefore emphasize realism over controlled censoring manipulation, while also requiring substantial data and computation.

When the goal is methodological evaluation—e.g., robustness and bias analyses across censoring regimes—it is highly desirable to have data where the censoring rate can be varied precisely *without* altering the underlying censoring mechanism. In particular, changing the censoring proportion should not wash out (or artificially create) covariate-driven censoring patterns present in the original data. We therefore seek a framework that enables fine-grained, controlled manipulation of the global censoring proportion while remaining faithful to the data-generating censoring process. In this work, we focus on a semi-synthetic setting: we keep covariates and event times fixed, and manipulate only the censoring process to reach a user-specified target rate.

To this end, we make the following contributions:

- **Controlled semi-synthetic censoring.** We introduce a protocol that starts from the initially uncensored subjects of a real survival dataset and generates multiple semi-synthetic versions with a user-specified censoring rate (e.g., 10–90%) within this subgroup, while keeping covariates and event times fixed. This enables controlled benchmarking across a wide range of censoring regimes.
- **Realism without breaking mechanisms.** We propose a modular censoring generator that combines (i) risk-aware selection of which individuals to censor (preserving covariate-censoring patterns), and (ii) data-driven assignment of censoring times that matches an empirical reference distribution. This design aims to modify the global censoring rate without washing out, or artificially creating, covariate-driven censoring effects.
- **Empirical validation.** Across six real datasets and multiple target rates, the simulated censoring distribution better matches an empirical reference (via Wasserstein distance to a Kaplan–Meier-based target); on controlled synthetic settings,

the method better preserves event–censoring dependence (measured by Kendall’s τ), outperforming baselines and variants.

We further illustrate how this controlled setup can be used to run robustness and bias analyses for survival evaluation metrics under systematically varied censoring.

2 Basic Concepts of Survival Analysis

Survival Data and Censoring. A standard survival dataset comprises N observations, each represented by a triplet (X_i, t_i, δ_i) , where $X_i \in \mathbb{R}^d$ is a vector of d covariates for the i -th subject, $t_i \in \mathbb{R}^+$ is the observed time, and $\delta_i \in \{0, 1\}$ is an event indicator. Formally, $t_i = \min(T_i, C_i)$ and $\delta_i = \mathbb{1}[T_i \leq C_i]$, where T_i is the *event time* and C_i is the *censoring time*. If $\delta_i = 1$, the event occurred at time $t_i = T_i$; if $\delta_i = 0$, the event was not observed, and $t_i = C_i$.

Censoring is said to be *left*, *right*, or *interval*, depending on whether the event occurs before, after, or within a known time interval. Right censoring being the most common in survival analysis, this work focuses exclusively on this case.

Key Functions in Survival Analysis. A central goal in survival analysis is to describe the distribution of time to event for a population or for individuals with specific covariates. The primary function used is the *survival function*: $S(t | x) = \Pr(T > t | x)$

For simplicity, the condition on x is often omitted in notation, though one can distinguish between population-level and individual-level survival as needed.

Another key concept is the *hazard function*, capturing the instantaneous rate at which events occur given that the subject has survived up to time t :

$$h(t | x) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t \leq T < t + \Delta t \mid T \geq t, x)}{\Delta t}.$$

3 Proposed Approach

Our proposed method for generating synthetic censored data follows two main steps: first, we select a subset of initially uncensored subjects to be censored in order to achieve a predefined global censoring rate within this subgroup; second, we assign synthetic censoring times to this subset. Algorithm 1 summarizes the full procedure.

3.1 Selection of Subjects to Censor

This stage is a critical component of the approach, ensuring a structured and realistic selection process rather than a purely random selection. It follows three sub-steps:

Estimating the Censoring Distribution. To model the censoring mechanism, we invert the censoring indicator: $\delta'_i = 1 - \delta_i$. We then fit a survival analysis model treating censoring as the event of interest in this reversed problem, and estimate the *distribution of censoring times*. This standard “reverse-survival” construction provides a flexible, data-driven approximation of the censoring process. The resulting survival function of censoring for an individual i , conditional on covariates X_i (if predicted at the individual level), is defined as:

$$\hat{S}_{C,i}(t) = \hat{P}(C_i > t \mid X_i). \quad (1)$$

From this, we compute the associated cumulative distribution function (CDF) of censoring:

$$\hat{F}_{C,i}(t) = 1 - \hat{S}_{C,i}(t) = \hat{P}(C_i \leq t \mid X_i). \quad (2)$$

Calculating Estimated Censoring Probabilities. For each subject i who was initially uncensored ($\delta_i = 1$), we compute the probability of being censored before their event time T_i :

$$\hat{P}_{C,i} = \hat{F}_{C,i}(T_i). \quad (3)$$

Probability-Based Selection of Subjects. Let M_{event} be the number of subjects with observed events ($\delta = 1$) and p the target censoring rate. We then define $m = \lfloor p \cdot M_{\text{event}} \rfloor$ and select, without replacement, exactly m individuals from this set using sampling weights $\hat{P}_{C,i}$.

However, weighted sampling without replacement induces *selection bias*, since inclusion probabilities drift from the original $\hat{P}_{C,i}$ as individuals are removed. This effect can substantially distort the intended censoring-risk profile of the selected subset, so we correct it using an iterative empirical reweighting scheme (Appendix A, Algorithm 2) inspired by unequal-probability survey sampling designs.

3.2 Generation of Synthetic Censoring Times

After selecting these m individuals, we assign a synthetic censoring time $\tilde{C}_i$ to each of them. To ensure that censoring occurs before their event time T_i , we use the *subject-specific truncated CDF*:

$$F_i^*(t) = \begin{cases} 0 & t < 0, \\ \frac{\hat{F}_{C,i}(t)}{\hat{F}_{C,i}(T_i)} & 0 \leq t < T_i, \\ 1 & t \geq T_i. \end{cases} \quad (4)$$

A censoring time is then sampled by inverse transform. More precisely, we draw

$$u \sim U(0, \hat{P}_{C,i}),$$

where $\hat{P}_{C,i} = \hat{F}_{C,i}(T_i)$ is the individual probability of being censored before their true event time. The censoring time is then obtained as:

$$\tilde{C}_i = \inf\{t : F_i^*(t) \geq u\}. \quad (5)$$

This construction guarantees that $\tilde{C}_i < T_i$ almost surely, while preserving the individual-specific censoring distribution up to T_i .

4 Experiments

4.1 Baseline Approaches

Although several existing methods in the literature generate fully synthetic survival datasets *from scratch*, our goal differs in that we seek to *modify* an already existing dataset to achieve a chosen censoring rate. This allows us to obtain more realistic generated dataset with controlled censoring properties. To the best of our knowledge, no previous work directly addresses this semi-synthetic scenario, therefore, we introduce three baseline approaches for comparison. Furthermore, to better understand the contribution of each component of our proposed approach, we introduce two additional ablation variants.

Algorithm 1 Semi-Synthetic Censoring Generation

Input: Dataset $\mathcal{D} = \{(X_i, T_i, \delta_i)\}_{i=1}^N$, target censoring rate $p \in (0, 1)$; survival model g for censoring; time grid $\mathcal{T}$ (e.g., unique observed times); random seed s .

Output: Updated dataset with a subset of initially uncensored subjects relabeled as censored, and assigned synthetic censoring times $\tilde{C}_i$.

1. Fit a censoring model and compute individual censoring risks

Define the censoring indicator $\delta'_i = 1 - \delta_i$ and fit g on $\{(X_i, T_i, \delta'_i)\}_{i=1}^N$.

Let $\mathcal{I} = \{i : \delta_i = 1\}$ be the set of initially uncensored subjects and $M_{\text{event}} = |\mathcal{I}|$.

for each $i \in \mathcal{I}$ **do**

Estimate $\hat{S}_{C,i}(t \mid X_i)$ on $t \in \mathcal{T}$, and $\hat{F}_{C,i}(t) = 1 - \hat{S}_{C,i}(t)$.

Compute $\hat{P}_{C,i} = \hat{F}_{C,i}(T_i)$.

end for

2. Select exactly m subjects to censor (bias-corrected WS-WoR)

Set $m = \lfloor p \cdot M_{\text{event}} \rfloor$.

Obtain corrected sampling weights w_i from $\{\hat{P}_{C,i}\}_{i \in \mathcal{I}}$ using Algorithm 2.

Draw a subset $\mathcal{J} \subset \mathcal{I}$ with $|\mathcal{J}| = m$ by weighted sampling without replacement using w_i (seed s).

3. Assign synthetic censoring times by truncated inverse transform

for each $i \in \mathcal{J}$ **do**

Sample $u_i \sim U(0, \hat{P}_{C,i})$.

Set $\tilde{C}_i = \hat{F}_{C,i}^{-1}(u_i)$ (inverse w.r.t. t , evaluated on $\mathcal{T}$ with interpolation if needed).

Update observation: $T_i \leftarrow \tilde{C}_i$, $\delta_i \leftarrow 0$.

end for

Return Updated dataset.

Baseline 1: Random selection of censored individuals and censoring times. This baseline mimics the two-step structure of our method but discards covariate information and the empirical censoring distribution. First, among individuals who originally experienced the event ($\delta_i = 1$), a subset is uniformly sampled to match the target censoring rate. Then, for each selected subject i , a censoring time C_i is drawn uniformly from the interval $[0, T_i]$.

Baseline 2: Parametric Exponential censoring (rate matching). We generate censoring times for individuals with $\delta_i = 1$ from an exponential distribution, $\tilde{C}_i \sim \text{Exp}(\theta)$. To ensure valid censoring, we retain only draws satisfying $\tilde{C}_i < T_i$. The parameter θ is adjusted via an iterative rate-matching procedure so that the achieved censoring proportion matches the target within a small tolerance. Implementation details and tuning are in Appendix B.1; achieved rates (mean $\pm$ std) are in Appendix B.3.

Baseline 3: Parametric Weibull censoring (rate matching). We also consider a Weibull baseline, $\tilde{C}_i \sim \text{Weibull}(k, \lambda)$, which generalizes the exponential case and is frequently used in survival modeling. We fix the shape parameter k and tune the scale λ to

match the target censoring rate, again keeping only draws with $\tilde{C}_i < T_i$. As with the exponential baseline, implementation details and tuning are in Appendix B.2; achieved rates (mean $\pm$ std) are in Appendix B.3.

Adaptive Time Assignment: Random selection with CDF-based assignment. This variant performs a random selection of the d subjects to censor as Baseline 1, but assigns their $\tilde{C}_i$ by sampling from the inverse of the estimated CDF of censoring times, as the second step of our approach.

Adaptive Sampling: Intelligent selection with uniform assignment. This approach mirrors the first step of our method by selecting censored individuals using our estimated censoring probabilities, but with $\tilde{C}_i \sim U(0, T_i)$.

4.2 Datasets

We evaluated our approach on six widely used survival datasets from the medical domain, spanning diverse sample sizes, feature counts, and censoring rates. They include cancer cohorts (GBMLGG, METABRIC, NACD) [10, 11] and critical care studies (SUPPORT, PBC) [13, 14]. Table 1 summarizes their characteristics.

Table 1: Summary of the six survival analysis datasets used. Columns: (1) initial censorship rate, (2) number of subjects, (3) number of features, (4) maximum event time.

Dataset	% Cens.	# Subj.	# Feat.	Max t
SUPPORT	31.9	9,105	31	2,029
NACD	36.5	2,395	48	84.3
GBMLGG	44.2	1,102	13	6,423
METABRIC	55.2	1,981	79	9,218
PBC	61.5	418	17	4,795
FLCHAIN	72.5	7,871	23	5,215

In addition to real-world datasets, we also generate synthetic data to evaluate how well different methods preserve the dependency between event times T and censoring times C . Real datasets do not provide both T and C for each individual and do not guarantee a controlled dependence structure, since only $\min(T, C)$ and the event indicator are observed. Synthetic generation overcomes this limitation by ensuring full observability of both variables under a prescribed dependency.

We therefore construct six synthetic trials in which both T and C are explicitly generated. For each trial, covariates T are first sampled, together with independent random variables $U, V \sim \mathcal{U}(0, 1)$. These are then transformed into event times T and censoring times C through Weibull AFT models with covariate effects, so that T and C share the same covariate basis but differ through their respective coefficient vectors. This design induces a controlled dependence between T and C via the correlation between the linear predictors linked to β_T and β_C .

To ensure comparability across trials, the overall censoring rate is calibrated to 50% in all synthetic datasets. Across the six trials, we vary β_T and β_C to induce different levels of dependence between T and C while keeping the censoring proportion fixed. Further

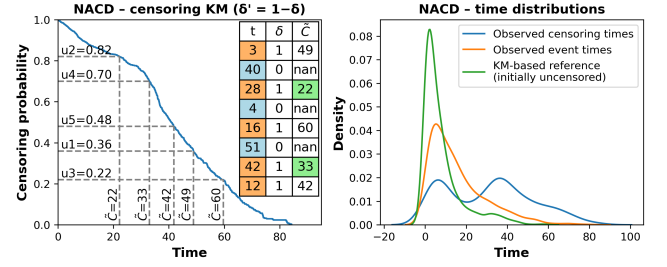


Figure 1: (Left panel) Kaplan-Meier estimate of the censoring survival function obtained from inverted indicators. The table illustrates the initial event (orange) and censor (blue) times. Sampling from its generalized inverse defines a non-parametric censoring mechanism at the population level and provides sampled censored times $\tilde{C}$ for all initially uncensored events. Only sampled censored times lower than their respective event times are kept for the reference distribution (green). (Right panel) Empirical distribution of observed event and censoring times in the initial dataset and the KM-based reference censoring times for originally uncensored subjects, illustrating the systematic left-shift induced by the constraint $C < T$. Example illustrated from the NACD dataset.

implementation details on the generation of these datasets are provided in Appendix C.

4.3 Creation and role of a Reference Distribution

Since our approach only generates censoring times for subjects with $\delta_i = 1$, the distribution of the generated censored times can only be left-shifted from the original event times distribution and cannot be compared to the original censoring distribution (Figure 1). Intuitively, evaluating a censoring generator requires a counterfactual benchmark: for each originally uncensored subject, how and when would censoring have occurred under the same mechanism observed in the initial dataset? The empirical censoring distribution of the full dataset cannot answer this directly, because each subject reveals at most one of (T, C) and the observed censoring times are already truncated by competing events. We propose a Kaplan-Meier-based reference to approximate this counterfactual censoring distribution under the same population-level mechanism, providing a reproducible target against which different generators can be compared via distributional distances.

Kaplan-Meier-based construction. The reference distribution is constructed by fitting a Kaplan-Meier (KM) estimator using the inverted censoring indicator ($\delta' = 1 - \delta$) across the entire population. This assumes that censoring is non-informative at the population level, i.e., independent of event times once covariates are ignored. We view this as a pragmatic approximation that yields a simple, reproducible benchmark distribution, rather than as a fully realistic generative model of censoring. More sophisticated alternatives such as stratified KM or covariate-informed mixtures could better capture heterogeneity in censoring risks, but for simplicity and reproducibility we rely on the plain KM estimator in all experiments.

Sampling procedure. To generate synthetic censoring times from the KM estimator, we use the generalized inverse of the stepwise CDF (Figure 1). Given the KM estimate $\hat{F}_C(t)$, we draw $u \sim U(0, 1)$ and set

$$\tilde{C} = \inf\{t : \hat{F}_C(t) \geq u\}.$$

Because the KM curve is a step function, we adopt the left-continuous inverse convention; when u falls between two jumps of $\hat{F}_C$, we interpolate linearly, and ties are broken at random. To enforce the natural constraint $\tilde{C}_i < T_i$, only valid times are retained, which shifts the distribution toward earlier values (Figure 1).

The KM-based reference is built once from the original data and used uniformly across conditions. Its censoring rate is not arbitrarily fixed but reflects, by construction, the proportion that initially uncensored individuals would exhibit if censored under the same mechanism. Thus, it is not intended to replicate the full-cohort distribution but to approximate the counterfactual censoring of this subgroup.

While the joint distribution of (T, C) is not identifiable from right-censored data, our KM-based reference offers a non-parametric, reproducible approximation of the underlying mechanism at the population level, which we use as a fixed evaluation anchor across censoring rates and methods.

4.4 Experimental Setup

For each of the six datasets, we applied our approach, baseline methods, and ablation variants to generate synthetic censoring times under varying rates. Starting from individuals with $\delta_i = 1$, we created ten modified versions per dataset, targeting censoring rates from 10% to 90% (step 10%). Our approach (and the *Adaptive Time Assignment* variant) requires an estimator of the censoring distribution (1). To assess robustness to this choice and to simulate different censoring mechanisms, we consider three censoring estimators: (i) a marginal Kaplan–Meier (KM) estimator, to simulated independent censoring, (ii) a Cox proportional hazards model, and (iii) a nonparametric Random Survival Forest (RSF) estimator to simulate covariate-dependent censoring.

KM-based censoring estimator. To simulate a marginally independent censoring scheme, we estimate the censoring distribution at the population level using Kaplan–Meier on the reversed indicator. This yields a single marginal survival curve $\hat{S}_C(t)$ (and CDF $\hat{F}_C(t)$) shared by all individuals, i.e., $\hat{F}_{C,i}(t) \equiv \hat{F}_C(t)$ for every i .

Cox-based censoring estimator. We fit a Cox proportional hazards model¹ on the “reversed” censoring indicator (censoring treated as the event) so as to estimate covariate-dependent censoring risk. Parameters $\hat{\beta}$ are obtained via partial likelihood maximization with Efron’s method for ties (default in `lifelines`). The estimated hazard function for censoring is

$$\hat{h}_C(t | X_i) = \hat{h}_{C,0}(t) \exp(\hat{\beta}^T X_i), \quad (6)$$

where $\hat{h}_C(t | X_i)$ denotes the instantaneous censoring rate at time t for covariates X_i , $\hat{h}_{C,0}(t)$ is the baseline hazard, and $\hat{\beta}$ are regression coefficients.

¹Implemented with `lifelines` [5]

RSF-based censoring estimator. Finally, we use a Random Survival Forest to capture potentially nonlinear and interaction effects in censoring risk while remaining nonparametric. We treat censoring as the event as above and obtain individual-specific estimates $\hat{S}_{C,i}(t)$ and $\hat{F}_{C,i}(t)$. Since the RSF is used only as a censoring estimator, we favor stable and smooth estimates using conservative settings: `n_estimators=300`, `max_depth=5`, `min_samples_split=40`, `min_samples_leaf=20`, `max_features=sqrt`, `random_state=123`, and `n_jobs=-1`.

In practice, since the corresponding survival and cumulative distribution functions are estimated as step functions, two implementation choices are required when drawing censoring times by inverse transform: (i) we adopt the left-continuous inverse convention, and (ii) when the sampled u falls between two jumps of $\hat{F}_{C,i}$, we interpolate linearly between the closest jumps to obtain a continuous approximation.

To assess how well the censoring times generated by our method and the baselines align with the reference censoring pattern, we calculate the Wasserstein distance [18] between the reference distribution and the empirical distribution of simulated censoring times at each target censoring rate and for each dataset. Wasserstein distance is a well-known distance between distributions and is easy to compute when these distributions are 1D as in our case. To provide a robust assessment, we generated 10 independent synthetic datasets for each target censoring rate and for each dataset. We thus computed the mean and standard deviation of the Wasserstein distances² across the 10 generated datasets for each target censoring rate.

In addition to evaluating the realism of the generated censoring times, we assessed the ability of our method to preserve the natural dependency between event and censoring times when generating covariate-dependent censoring with Cox-based and RSF-based censoring estimators, an important feature often present in real-world data. Each method tested to generate censored data was applied at different censoring rates to all synthetic datasets presented in Section *Experiments*. For each newly generated dataset, a Clayton copula was then fitted to the observed (T, C) pairs to estimate their Kendall’s τ . As the observed (T, C) pairs correspond to the individual with generated censor time with $C < T$, we estimated the reference Kendall’s τ for the population with $C < T$ in the initial synthetic dataset (before generating censor times).

5 Simulation Results

5.1 Qualitative Evaluation

We assess the generated censoring times with kernel density estimation (KDE). Figure 2 shows the densities for each dataset, comparing our method (with the Cox-based censoring estimator) with three baselines and two ablation variants. The target censoring rate matches the expected rate from the reference distribution.

The proposed approach (KM, Cox, RSF) produces censoring time distributions that most closely resemble the empirical reference distribution. Specifically, it maintains the general shape and density concentration of the reference KDE, capturing both the main modes and tail behavior of the censoring times. In contrast, the

²Computed using Python `scipy.stats` package.

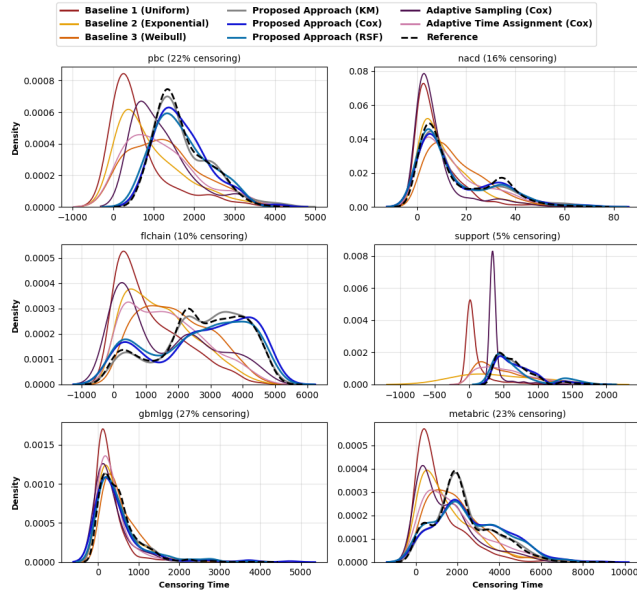


Figure 2: KDE plots comparing the distribution of generated censoring times across methods. For each dataset, simulations are run at its expected censoring rate defined by the reference distribution (percentage indicated in parentheses). For each method, the KDE is computed from the pooled censoring times across replications (i.e., aggregated to show a single curve per method), enabling a clearer comparison of the resulting censoring-time distributions.

baseline methods exhibit noticeable distortions. *Baseline 1* produces systematically biased distributions, heavily skewed towards early censoring, which prevents it from capturing the tail behavior observed in the reference KDE. *Baseline 2 and 3* is less skewed than the uniform approach but still struggles to approximate the reference distribution. This limitation is particularly evident in datasets *Support*, *Flchain*, *Metabric*, and *NACD*, where the reference KDE significantly deviates from a simple exponential decay.

Adaptive Time Assignment and *Adaptive Sampling* reduce some of these distortions by incorporating either a more realistic assignment of censoring times or a more informed selection mechanism, respectively. As a result, their generated distributions are moderately closer to the reference than those of *Baseline 1* and *2*. However, both remain unable to fully replicate the structure of the reference KDE, particularly in capturing the multimodality and tail behavior observed in real censoring patterns.

These results emphasize the ability of our method to generate censoring times that better preserve the structure of the censoring times distribution.

5.2 Quantitative Evaluation

As shown in Figure 3, the evolution of the Wasserstein distance as a function of the target censoring rate provides further insights into the behavior of each method. *Our approach consistently yields lower or comparable Wasserstein distances across the tested censoring*

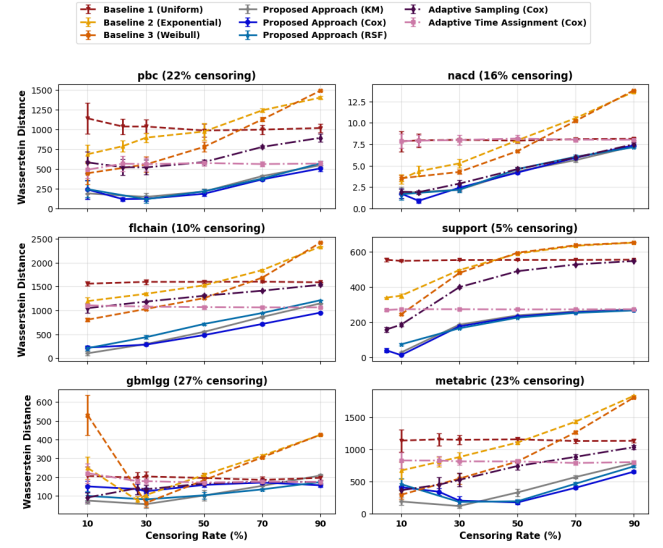


Figure 3: Evolution of the Wasserstein distance between the generated censoring distributions and the reference distribution across varying censoring rates for six datasets. Each curve represents a different method and the error bars indicate the standard deviation over ten simulations. Lower distances suggest a better match with the reference distribution.

rates compared to baselines. This conclusion holds similarly for all three alternative instantiations of our framework, which show very similar results, indicating that the distributional matching results are largely robust to the choice of censoring mechanism and censoring estimator. For our method, the distance generally reaches a minimum around the dataset’s expected censoring rate (though not necessarily at a single exact value) and increases as we deviate from this point. This behavior is expected, as the reference distribution is intrinsically linked to the dataset’s initial censoring structure. Modifying the dataset to achieve significantly higher or lower censoring rates inevitably alters this structure, leading to an increase in Wasserstein distance. However, even when deviating from the expected censoring rate, our approach exhibits a lower increase in distance compared to the baselines, demonstrating its robustness across different censoring conditions.

In contrast, the remaining five methods follow distinct trends. *Baseline 1* yields the highest Wasserstein distances at lower censoring rates. However, at higher censoring rates (above approximately 60%), it presents lower Wasserstein distances than *Baseline 2* and *Baseline 3*. The distance for *Baseline 1* remains relatively stable across censoring levels, a behavior that appears to stem primarily from the random selection of individuals to be censored rather than from the random choice of censoring time distribution, as the same behavior is observed for *Adaptive Time Assignment*. Since this process does not depend on the overall censoring rate, the distribution of generated censoring times remains largely unchanged, leading to a nearly constant Wasserstein distance. As illustrated in Figure 4, the density curves for *Baseline 1* appear similar across two

extremely different censoring rates, highlighting the insensitivity of this method to variations in censoring levels.

Baseline 2 (Exponential), despite presenting a minimum around the expected censoring rate, experiences a pronounced increase in distance when increasing the censoring rate. This effect is particularly severe at higher censoring rates, where the distance grows substantially. This behavior arises from the nature of the exponential distribution: to achieve a high censoring rate, the optimal parameter must be small, which results in a strong leftward shift of the censoring time distribution. This shift is evident in Figure 4, where for a 90% censoring rate, *Baseline 2* produces a highly skewed density with a sharp peak concentrated near zero. Such a distribution significantly diverges from the reference. *Baseline 3* (Weibull) is slightly more flexible and can yield marginally smaller distances than the exponential baseline at moderate rates, but it exhibits a similar deterioration as the censoring rate increases: rate matching at high censoring levels concentrates censoring times near zero, leading to increasingly poor alignment with the reference. This sharp deterioration highlights the limitations of using a parametric approach when the target censoring rate deviates from the dataset's original structure.

Adaptive Time Assignment exhibits a behavior similar to *Baseline 1*, with Wasserstein distances that remain relatively stable across different censoring rates. This stability arises from the fact that, like *Baseline 1*, the censoring time distribution remains fixed regardless of the censoring rate. However, due to the informed time assignment, it achieves better fidelity than *Baseline 1*, while still underperforming compared to our method.

Interestingly, *Adaptive Sampling* shows good performance on certain datasets, particularly *NACD* and *GBMLGG*. On the *GBMLGG* dataset, for instance, it even outperforms our method at low censoring rates and achieves comparable distances at higher rates. For other datasets, however, its performance is less favorable, showing consistently higher Wasserstein distances than our method, though it still remains superior to all three baselines.

These results highlight not only the robustness of our approach across diverse datasets and censoring scenarios, but also the importance of combining the two core components of our method: intelligent subject selection and informed censoring time assignment. Each component alone offers improvements over existing baselines, yet neither is sufficient on its own.

5.3 Dependency Structure Preservation

Figure 5 presents the results from the synthetic datasets. The left panel shows the evolution of Kendall's τ across censoring rates for a single trial, while the right panel reports the values obtained for all six trials (with all censoring rates pooled).

Our approach. With the Cox-based censoring estimator, our method yields τ values that remain stable and closely aligned with the reference (red dashed line), demonstrating its ability to preserve the original dependency structure under varying censoring rates—a property that the parametric and random baselines systematically fail to achieve. This property generalizes consistently across trials (with different initial τ), where our method also provides the closest match to the reference. Replacing Cox with a Random Survival Forest leads to a similar conclusion although RSF-based censoring

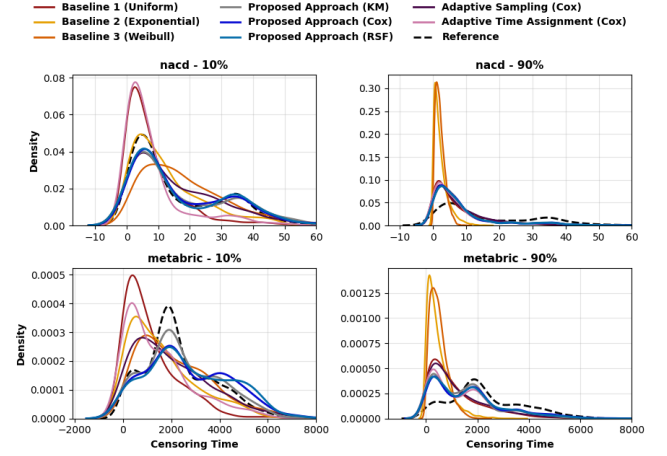


Figure 4: KDE plots comparing the distribution of generated censoring times across different methods at two censoring rates (10% and 90%) for the *nacd* and *metabric* datasets. The full set of plots across all target rates and datasets is provided in Appendix D.

estimator struggles to fully capture the event-censoring dependency when the dependency increases, potentially because of overfitting issues with RSF. However, the results indicate that the dependency preservation does not hinge on a specific parametric censoring model. In contrast, using a marginal Kaplan–Meier estimator yields systematically lower τ values. This behavior is expected because KM estimates a single population-level censoring distribution shared by all individuals, simulating a covariate-independent censoring mechanism.

In contrast, *Baseline 2* (Exponential) systematically underestimates the dependency, already at low censoring rates and increasingly so as the censoring rate grows. At high censoring, the estimated τ approaches zero, reflecting near independence between T and C . This degradation stems from the parametric nature of the exponential model: to reach high censoring levels it generates extremely small censoring times, which destroys the correlation. *Baseline 3* (Weibull) exhibits a similar behavior: although it is more flexible than the exponential family, matching extreme censoring rates still concentrates censoring times near zero, weakening the induced T – C association and driving τ downward.

Baseline 1 (Uniform) produces more stable τ values, but these remain systematically distant from the reference. This arises from its construction: censoring times are drawn uniformly between 0 and T , inducing an artificial but consistent dependency. Changing the censoring rate only modifies the proportion of censored subjects, not the underlying mechanism, hence the stability.

These effects also appear in the two variants of our method using Cox-based censoring estimator. The *Adaptive Sampling* variant, which generates censoring times as in the uniform baseline, inherits the same artificial dependency—probability-based selection of censored individuals does not restore the true structure. By contrast, the *Adaptive Time Assignment* variant assigns times via inverse sampling from the estimated censoring distribution,

and thus preserves the natural dependency, closely matching the reference.

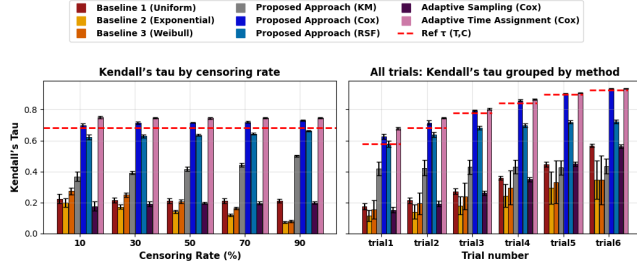


Figure 5: Left: Kendall's τ between event and censoring times by censoring rate (trial 1). Right: Kendall's τ grouped by method across trials (trials 1–6; all censoring rates pooled). Red dashed lines show the reference τ from the original synthetic dataset. Bars report the mean and error bars indicate $\pm$ one standard deviation across repetitions.

5.4 Use Case: Diagnosing Evaluation Gaps Under Controlled Censoring

A direct possible application of our framework is to stress-test evaluation metrics under systematically varied censoring regimes, while keeping covariates and event times fixed. Importantly, our generator can instantiate different censoring mechanisms through the choice of the censoring estimator used in Section 4.4: using a covariate-dependent Cox model yields a covariate-dependent censoring (CDC), whereas using a marginal Kaplan–Meier estimator yields a marginally independent censoring (IC) scheme.

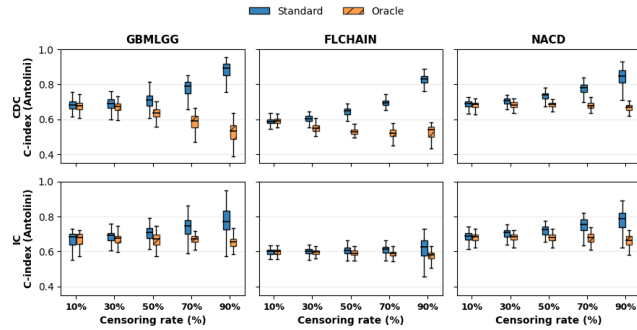


Figure 6: Metric-bias use case under controlled censoring. Antolini C-index computed on test folds with the standard right-censored definition (STANDARD) and with an oracle computation using true event times for synthetically censored cases (ORACLE). Results are shown across target censoring rates under CDC (top) and IC (bottom) for three datasets.

Figure 6 illustrates this use case on Antolini's time-dependent concordance [1]. For each dataset, target censoring rate, and mechanism (CDC, IC), we perform nested cross-validation (details in

Appendix E) and evaluate a representative set of survival models (CoxPH, Weibull AFT, RSF, DeepHit, and NMTLR)³. We report the metric computed in two ways: (i) STANDARD, using the right-censored test outcomes (Y, δ), and (ii) ORACLE, recomputing the same metric after replacing Y by the true event time T for synthetically censored individuals (a diagnostic reference enabled by our controlled setup).

Across datasets, the STANDARD concordance is systematically higher than ORACLE, and the gap widens as the censoring rate increases, with a markedly larger divergence under CDC than under IC. This controlled setting enables systematic quantification of censoring-induced metric bias, disentangling the impact of the censoring *rate* from that of the censoring *mechanism*.

While we illustrate the idea with Antolini's C-index, the same controlled setup can be used to probe other survival metrics and their variants, helping detect censoring-induced instabilities that threaten reliable benchmarking.

6 Discussion and Future Work

The ability to generate datasets with controlled censoring rates provides a valuable experimental framework to study censoring-related biases, including their impact on parameter estimation and evaluation metrics such as the concordance index or calibration.

A key strength of our method is its flexibility: using KM-based censoring estimator allows to simulate independent censoring whereas using individual based estimators such as Cox or RSF allows to simulate dependent censoring. Our distributional conclusions (Wasserstein-based matching) are largely unchanged across estimators, whereas the dependence signal (Kendall's τ) is preserved with covariate-dependent estimators (Cox/RSF) and is naturally reduced under KM as expected. Importantly, the censoring model is trained only to estimate the censoring mechanism, not the event process, so even if Cox or RSF are later used for survival prediction, the two roles remain distinct and no predictive overfitting is introduced.

Another advantage is simplicity and precision: our procedure achieves the exact target censoring rate by construction, without iterative parameter tuning, unlike parametric rate-matching baselines (exponential/Weibull), which provide approximate control on finite samples (Appendix B.3). This direct control is particularly beneficial on datasets such as *SUPPORT*, where the exponential baseline can exhibit larger deviations from the target at extreme rates (Appendix B.3).

Beyond controlling the censoring rate, our approach also preserves the natural dependency between event and censoring times when individual based estimators are used, a feature often present in real data but rarely captured by baselines (see Section 5.3).

This design choice enables exploration of both higher and lower censoring levels compared to the original dataset. However, the framework is not limited to generating *semi-synthetic* data from initially uncensored subjects. If needed, originally censored individuals could also be retained to allow more gradual adjustments. One might be careful with datasets with initial very low or very high censoring rates, as very high censoring might leave too few uncensored cases, while very low censoring reduces reliability of

³CoxPH and Weibull AFT are implemented using *lifelines*, RSF using *scikit-survival*, and DeepHit and NMTLR using *pycox*.

censoring function estimation. Such potential issues could be mitigated by incorporating censored cases or refining the estimation procedure. Beyond right censoring, the same principles could be extended to more complex schemes such as truncation or recurrent/multiple events, provided the censoring model and sampling procedure are adapted accordingly.

Finally, access to both true event times and synthetic censoring times opens new perspectives for studying biases of evaluation metrics under different censoring regimes. This dual information allows computing metrics in parallel with and without censoring, extending recent analyses [6, 20] and offering deeper insights into the robustness of survival models and metrics (see Section 5.4). In practice, our generator can be plugged into existing survival-analysis pipelines to augment public benchmarks with systematic censoring sweeps, without requiring any modification of downstream models. We hope this will encourage the community to report performance not just at a single, fixed censoring rate, but along full censoring profiles, making robustness to censoring a first-class evaluation criterion.

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A Bias in Weighted Sampling

Weighted sampling without replacement does not preserve the initial target weights exactly, because selection probabilities are updated after each draw as previously selected individuals are removed from the pool. Consequently, empirical inclusion probabilities may deviate from the intended weights, motivating the use of an iterative correction procedure to reduce this discrepancy.

To facilitate reproducibility, we provide below the empirical iterative correction used in our experiments:

The algorithm starts from normalized weights $w_i^{(0)} = \hat{P}_{C,i} / \sum_j \hat{P}_{C,j}$ and repeatedly simulates K weighted samples of size m . At each iteration, the empirical inclusion probabilities $\hat{\pi}_{i,t}$ are compared to the target probabilities $\hat{P}_{C,i}$, and the weights are updated as $w_{i,t} = w_{i,t-1}(1 - d_{i,t})$ with $d_{i,t} = (\hat{\pi}_{i,t} - \hat{P}_{C,i}) / \hat{P}_{C,i}$, followed by normalization to ensure nonnegativity and $\sum_i w_{i,t} = 1$.

This procedure reduces the gap between empirical inclusion probabilities π_i and targets $\hat{P}_{C,i}$ under $\sum_i \pi_i = m$. The update rule is heuristic but simple, effective, and avoids the bias of weighted sampling without replacement. Gradient-based optimization is theoretically possible but impractical, since inclusion probabilities change at each draw and lack a closed-form gradient. Other principled methods (e.g., Conditional Poisson, Sampford) exist, but this correction is preferred for its simplicity, reproducibility, and ease of use. Its complexity is $O(Q \cdot K \cdot m)$, with convergence usually within 10 iterations ($Q = 10$, $K = 1000$), under one minute on a standard workstation.

As shown in Figure 7, the naive use of raw probabilities leads to a clear mismatch between estimated censoring probabilities (x-axis) and empirical inclusion probabilities (black curve). After applying the iterative correction, the empirical inclusion probabilities (green

Algorithm 2 Iterative Bias Correction for Weighted Sampling Without Replacement

Input: Initial probabilities $\hat{P}_{C,i}$ for M individuals, target sample size m , iterations Q , repetitions per iteration K , threshold ϵ

Initialize: $w_{i,0} = \hat{P}_{C,i} / \sum_j \hat{P}_{C,j}$

for $t = 1$ to Q **do**

 Draw K samples of size m without replacement using weights $w_{i,t-1}$

 Calculate empirical inclusion probabilities $\hat{\pi}_{i,t}$ for each individual

 Compute deviation $d_{i,t} = (\hat{\pi}_{i,t} - \hat{P}_{C,i}) / \hat{P}_{C,i}$

 Update $w_{i,t} \leftarrow w_{i,t-1} \times (1 - d_{i,t})$

 Normalize $w_{i,t} \leftarrow w_{i,t} / \sum_j w_{j,t}$

If $\frac{1}{M} \sum_i |d_{i,t}| < \epsilon$: **stop**

end for

Output: Corrected weights $w_{i,t}$ and implied probabilities π_i

curve) closely match the target probabilities, demonstrating that the proposed procedure effectively eliminates the bias introduced by weighted sampling without replacement.

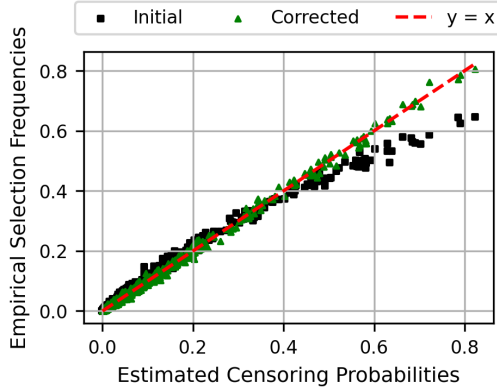


Figure 7: Selection bias correction for the FLCHAIN dataset. The x-axis shows the estimated censoring probabilities $\hat{P}_{C,i}$, and the y-axis the empirical inclusion probabilities over 1000 draws. Results are reported using raw weights (black) and after applying the iterative correction (green).

A.1 Alternative Design-Based Methods

Survey sampling offers several principled unequal-probability designs that target exact or nearly exact inclusion probabilities, including conditional Poisson sampling [7], which conditions independent Poisson draws on selecting exactly m units but requires solving non-linear constraints; Brewer’s *zpswor* algorithm [4], which provides a simpler approximation with nearly unbiased inclusion probabilities but may slightly deviate from the target sample size in finite populations; and Sampford’s method [24], which iteratively rescales weights to achieve prescribed inclusion probabilities but is computationally demanding for large M . Although theoretically attractive, these methods are impractical in our repeated simulation setting,

which requires many weighted draws. We therefore use the *empirical iterative correction* described in Algorithm 2 to directly reduce the gap between empirical and target inclusion probabilities.

B Parametric Baselines: Exponential and Weibull Rate Matching

B.1 Exponential Baseline: Parameter Tuning and Achieved Rates

Setup and notation. For each trial dataset, we first build a fully-observed subset by retaining only individuals with observed events in the original data, i.e., $\delta_i = 1$. For each such individual i , let $T_i > 0$ denote the original event time. We then generate a synthetic censoring time $\tilde{C}_i$ and define the observed time and indicator as $X_i = \min(T_i, \tilde{C}_i)$ and $\delta_i = \mathbb{1}\{T_i \leq \tilde{C}_i\}$.

Exponential censoring model. We draw $\tilde{C}_i$ from an exponential distribution, $\tilde{C}_i \sim \text{Exp}(\lambda)$, where $\lambda > 0$ is the *rate* parameter (mean $1/\lambda$). Given a target censoring rate $\rho \in (0, 1)$, our objective is to choose λ such that the *achieved* censoring proportion $\hat{\rho}(\lambda) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{\tilde{C}_i < T_i\}$ matches ρ within a tolerance ϵ (we use $\epsilon = 0.05$). Since $\hat{\rho}(\lambda)$ is monotone increasing in λ for fixed $\{T_i\}$, larger λ yields smaller censoring times and thus a larger censoring proportion. We therefore use a simple iterative *rate-matching* procedure starting from the heuristic initialization $\lambda_0 = 1/\bar{T}$, where $\bar{T}$ is the mean event time among $\delta_i = 1$ individuals.

Rate-matching procedure. Let $\Delta = \alpha\lambda_0$ be a fixed step size with $\alpha > 0$ (we use $\alpha = 0.1$). At each iteration, we sample $\tilde{C}_i \sim \text{Exp}(\lambda)$ for all i , compute $\hat{\rho}(\lambda) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{\tilde{C}_i < T_i\}$, and update λ according to the achieved rate. If $\hat{\rho}(\lambda) < \rho$, we increase the rate by setting $\lambda \leftarrow \lambda + \Delta$; otherwise, we decrease it by setting $\lambda \leftarrow \max(\lambda - \Delta, \lambda_{\min})$. The iterations continue until $|\hat{\rho}(\lambda) - \rho| \leq \epsilon$. Here $\lambda_{\min}$ is a small positive safety floor (e.g., 10^{-6}) to avoid non-positive rates.

Once λ is fixed for a given (trial, ρ), we generate R independent replications (we use $R = 10$) with different random seeds and report the achieved censoring rates in Appendix B.3.

B.2 Weibull Baseline: Parameter Tuning and Achieved Rates

Setup and notation. We follow the same construction as above: we restrict to individuals with $\delta_i = 1$, denote their event times by T_i , generate synthetic censoring times $\tilde{C}_i$.

Weibull censoring model. We consider a Weibull distribution parameterized by a shape $k > 0$ and a scale $\lambda > 0$, $\tilde{C}_i \sim \text{Weibull}(k, \lambda)$. In our implementation, k is fixed (we use $k = 1.5$) and we tune the scale λ to match a target censoring rate ρ . As in the exponential baseline, the achieved rate is $\hat{\rho}(\lambda) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{\tilde{C}_i < T_i\}$. We enforce $|\hat{\rho}(\lambda) - \rho| \leq \epsilon$ with a tighter tolerance $\epsilon = 0.01$.

Bracketed search (scale tuning). For fixed k and $\{T_i\}$, $\hat{\rho}(\lambda)$ is monotone *decreasing* in the Weibull *scale* λ (larger scale yields larger censoring times, hence less censoring). We therefore use a bracketed search initialized around $\lambda_0 = \bar{T}$. We create an interval $[\lambda_0/b, \lambda_0 \cdot b]$ (with $b = 10$), evaluate $\hat{\rho}(\lambda_{\text{low}})$ and $\hat{\rho}(\lambda_{\text{high}})$ at the two endpoints, and expand the bracket if needed. We then perform a bisection-like search to minimize the absolute error to the target rate. At each iteration $t = 1, \dots, T_{\max}$, we set $\lambda = (\lambda_{\text{low}} + \lambda_{\text{high}})/2$, sample

$\tilde{C}_i \sim \text{Weibull}(k, \lambda)$, compute $\hat{\rho}(\lambda)$, and update the best parameter value (λ^*, e^*) whenever $|\hat{\rho}(\lambda) - \rho| < e^*$. If $|\hat{\rho}(\lambda) - \rho| \leq \varepsilon$, the search stops. Otherwise, if $\hat{\rho}(\lambda) < \rho$, we set $\lambda_{\text{high}} \leftarrow \lambda$; otherwise, we set $\lambda_{\text{low}} \leftarrow \lambda$. Importantly, if the target cannot be reached within tolerance due to discreteness/finite-sample randomness, we keep the parameter value that yields the closest achieved rate (minimum $|\hat{\rho}(\lambda) - \rho|$).

Once the best scale λ^* is selected for a given (trial, ρ), we generate $R = 10$ replications using different random seeds and report achieved censoring rates in Section B.3.

B.3 Summary of Target vs. Achieved Censoring Rates

For each dataset and target censoring rate ρ , we report the achieved censoring proportion $\hat{\rho} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{\tilde{C}_i < T_i\}$ produced by the tuned parametric baselines described in Appendix B.1 and Appendix B.2. Tables 2 and 3 summarize the results for $\rho \in \{10, 30, 50, 70, 90\}\%$.

Table 2: Exponential baseline: achieved censoring rates (mean $\pm$ std over $R = 5$ replications) for targets $\rho \in \{10, 30, 50, 70, 90\}\%$.

Dataset	10%	30%	50%	70%	90%
PBC	10.56	30.43	50.31	70.19	90.68
NACD	9.53	29.50	50.66	69.71	89.62
FLCHAIN	9.05	30.98	50.51	70.54	89.24
SUPPORT	8.64	34.11	46.27	65.10	85.13
GBMLGG	9.43	29.43	49.27	69.27	89.43
METABRIC	9.91	29.73	50.34	69.93	89.98

Table 3: Weibull baseline: achieved censoring rates (mean $\pm$ std over $R = 5$ replications) for targets $\rho \in \{10, 30, 50, 70, 90\}\%$.

Dataset	10%	30%	50%	70%	90%
PBC	13.04 $\pm$ 2.22	34.53 $\pm$ 4.47	51.43 $\pm$ 2.82	69.19 $\pm$ 2.08	90.81 $\pm$ 1.29
NACD	10.09 $\pm$ 1.04	31.24 $\pm$ 0.92	50.55 $\pm$ 0.91	69.30 $\pm$ 0.35	89.90 $\pm$ 0.64
FLCHAIN	9.20 $\pm$ 0.90	30.80 $\pm$ 0.89	51.70 $\pm$ 0.75	70.16 $\pm$ 0.70	90.61 $\pm$ 0.33
SUPPORT	10.64 $\pm$ 0.15	30.13 $\pm$ 0.20	50.11 $\pm$ 0.64	69.49 $\pm$ 0.60	89.89 $\pm$ 0.35
GBMLGG	7.87 $\pm$ 1.64	30.99 $\pm$ 2.56	51.19 $\pm$ 1.01	70.41 $\pm$ 1.83	89.59 $\pm$ 0.74
METABRIC	9.35 $\pm$ 0.93	32.05 $\pm$ 1.33	49.26 $\pm$ 0.81	68.33 $\pm$ 0.62	90.54 $\pm$ 0.40

C Synthetic Data Generation for Dependency Evaluation

This section details the procedure used to generate the synthetic datasets described in Section 4.2, designed to evaluate whether censoring-generation methods preserve the dependency between event times T and censoring times C . Since real-world datasets do not provide both T and C for all individuals, we generate fully synthetic data where both times are observed and the dependency structure is controlled.

Overview. Each synthetic dataset (or “trial”) contains $N = 10,000$ individuals with covariates $X_i = (X_{i1}, X_{i2})$ independently sampled from $\mathcal{U}(0, 1)^2$. Event and censoring times are generated from Weibull AFT models with log-linear covariate effects:

$$T_i = \left(\frac{-\log(U_i)}{\exp(X_i^\top \beta_T)} \right)^{1/\rho_T}, \quad C_i = \left(\frac{-\log(V_i)}{s \cdot \exp(X_i^\top \beta_C)} \right)^{1/\rho_C}.$$

Here, $U_i, V_i \sim \mathcal{U}(0, 1)$ introduce stochastic variability, β_T and β_C control covariate effects on event and censoring times, ρ_T and ρ_C are Weibull shape parameters, and $s > 0$ scales the censoring model to regulate the overall censoring proportion.

Dependency Structure. The dependency between T and C is controlled through the linear predictors $X_i^\top \beta_T$ and $X_i^\top \beta_C$: stronger alignment between β_T and β_C induces stronger association between event and censoring times. In practice, both vectors are initialized identically, e.g., $\beta_T = \beta_C = (3.0, 3.0)$, then amplified after each trial as $\beta_T^{(k+1)} = 1.5 \beta_T^{(k)}$ and $\beta_C^{(k+1)} = 1.5 \beta_C^{(k)}$. This increases the variance of the linear predictors and produces datasets with progressively stronger dependency between T and C , while keeping the sample size, covariate distribution, and Weibull shapes fixed. The dependency therefore arises from shared covariate effects rather than from an imposed copula.

Calibration of the Censoring Rate. For each trial, the censoring rate π_{cens} is calibrated to a predefined target value (typically 50%) by adjusting the scaling parameter s until the observed censoring proportion $\pi_{\text{cens}} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}\{T_i > C_i\}$ matches the target within $\pm 2\%$. This iterative calibration converges rapidly (in fewer than ten iterations in practice) and keeps the censoring rate consistent across trials, independently of the dependency strength.

Empirical Verification. After each trial, we assess the dependency between T and C by refitting a Clayton copula to rank-based pseudo-observations of both variables and by computing the empirical Kendall’s $\tau(T, C)$. We compare this empirical τ with the theoretical τ implied by the fitted copula parameter, confirming that dependency increases monotonically with the correlation between β_T and β_C while the censoring rate remains fixed.

Outputs. Each synthetic dataset contains the true event time T_i , censoring time C_i , observed follow-up time Y_i , and event indicator δ_i , with $Y_i = \min(T_i, C_i)$ and $\delta_i = \mathbb{1}\{T_i \leq C_i\}$ equal to 1 if the event is observed and 0 otherwise. The covariate matrix X and an individual index are also included. Two CSV files are generated per trial: one with the complete data including both true times (T, C) , and one with only the observed triplets (Y, δ, X) used for survival model training. All trials share the same marginal Weibull parameters (ρ_T, ρ_C) but differ in the dependency strength determined by the correlation between β_T and β_C .

D Censoring Distributions Across All Rates and Datasets

Section 5.2 reported KDE comparisons for two datasets. We extend this analysis to all six datasets and all target censoring rates (10%, 30%, 50%, 70%, and 90%); see Figures 8 and 9. Overall, *Baseline 1* remains weakly affected by the target rate and misaligned with the reference distribution, while *Baseline 2* and the *Weibull* baseline

become increasingly left-skewed at high censoring rates. In contrast, all variants of *Our Approach* remain closer to the reference across datasets and rates, better preserving the distributional shape.

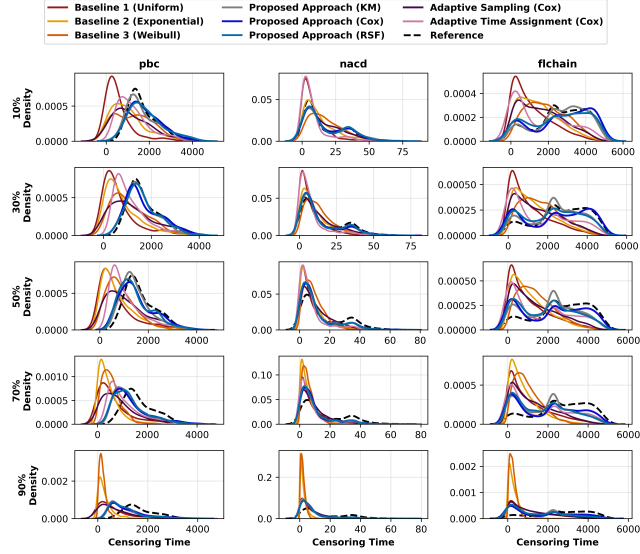


Figure 8: KDE plots comparing the distribution of generated censoring times across methods for the first three datasets (pbc, nacd, and flchain) and target censoring rates (10%, 30%, 50%, 70%, and 90%). The reference distribution (black, dashed) is contrasted with *Proposed Approach* (Cox/KM/RSF) and baseline generators (uniform, exponential, Weibull).

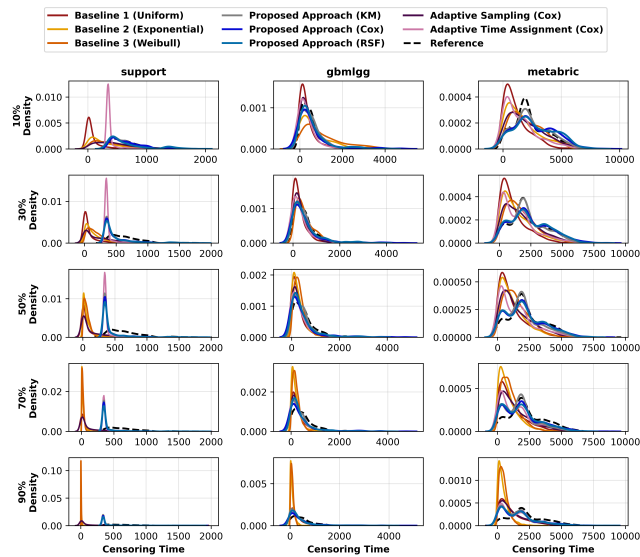


Figure 9: Same KDE comparison as in Figure 8, shown for the remaining datasets: support, gbmlgg, and metabric.

E Metric-bias use case: experimental protocol

This appendix provides additional details on the experimental pipeline used in the metric-bias use case reported in Figure 6. Across the six benchmark datasets, we apply our controlled censoring procedure to generate collections of datasets at multiple target censoring rates and under two censoring mechanisms. We instantiate IC using a nonparametric KM-based censoring reference distribution, and CDC using a covariate-dependent censoring model based on Cox regression. A key property of this setup is that the underlying event times are preserved through a `true_time` field for synthetically censored individuals, enabling a paired comparison between *STANDARD* and *ORACLE* evaluations while keeping the feature distribution and censoring design fixed.

E.1 Training protocol and hyperparameter selection

We evaluate five survival models: CoxPH, Weibull-AFT, RSF, DeepHit, and NMTLR.⁴ To avoid optimistic bias due to hyperparameter tuning, we use nested cross-validation for the three models requiring model selection (DEEPHIT, NMTLR, RSF): a 5-fold outer loop for evaluation and a 3-fold inner loop for hyperparameter selection. Hyperparameters are optimized with Optuna using a TPE sampler on the inner folds, targeting predictive discrimination on validation data. For neural models, the optimization objective is Harrell’s concordance computed from a cumulative-risk score derived from predicted survival curves.

For DEEPHIT, we tune dropout, learning rate, batch size, and the loss-specific parameters α and σ , which control the trade-off between the likelihood and ranking components. For NMTLR, we tune dropout, learning rate, and batch size. For RSF, we tune the number of trees, maximum depth, minimum samples per split, minimum samples per leaf, and the number of features considered at each split. In contrast, CoxPH and Weibull-AFT are fitted on each outer training split without hyperparameter optimization, beyond default regularization where applicable, and evaluated on the corresponding outer test split.

E.2 Replications and evaluation modes

For each dataset, censoring mechanism (IC vs. CDC), and target censoring rate, we generate $R = 5$ independent replications. For each replication, we run the full nested cross-validation pipeline: within each outer fold, hyperparameters are selected on the inner loop, then the model is retrained on the full outer training split and evaluated on the held-out outer test split.

All reported metrics are computed in two evaluation modes. *Standard* evaluation uses the observed right-censored test outcomes (y_i, δ_i) . *Oracle* evaluation replaces the test times by the fully observed event times T_i and sets event indicators to 1, using the stored `true_time` values for synthetically censored individuals. This produces paired metric values per fold and replication, enabling a controlled assessment of how censoring affects the metric when the underlying event times are known by construction.

⁴CoxPH and Weibull AFT are implemented using lifelines, RSF using scikit-survival, and DeepHit and NMTLR using pycox.